



SHABBIR ALAM

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ELECTRICAL ENGINEER

Highly motivated and results-oriented Electrical Project Engineer with over 12 years of experience in electrical construction, erection, testing, pre-commissioning, commissioning, start-up, and maintenance of electrical systems for power, industrial, infrastructure, and MEP projects. Possessing strong expertise in the installation, testing, troubleshooting, and energization of high-voltage, medium-voltage, and low-voltage electrical equipment, ensuring compliance with international standards, project specifications, and safety requirements.

Experienced in the execution of electrical works related to substations, transformer installations, switchgear systems, generators, PCC and MCC panels, cable laying, cable termination, cable testing, earthing systems, lighting, power distribution systems, and MEP electrical installations. Demonstrated ability to manage site activities, coordinate with multidisciplinary teams, contractors, consultants, and clients to ensure successful project completion within schedule and budget.

Holding a Master of Technology (M. Tech) in Electrical and Electronics Engineering with strong technical knowledge of power systems, electrical design, project execution, testing, commissioning, and operational reliability. Proven experience in electrical project management, construction supervision, quality control, technical documentation, and site coordination.

Strong commitment to workplace safety with recognized safety training certifications and practical experience as a Permit Receiver, ensuring strict compliance with HSE procedures, risk assessments, permit-to-work systems, lockout/tagout (LOTO) requirements, and safe work practices. Adept at identifying hazards, implementing corrective actions, and promoting a strong safety culture across project sites.

OVERSEAS EXPERIENCE :- 12 YEARS

COMPANY NAME-	AMERICAN TECHNO CONTROLS Inc. KSA
DESIGNATION-	ELECTRICAL ENGINEER
LOCATION	AL KHOBAR, EASTERN REGION
DURATION	2022 TO TILL NOW

PROJECT NAME	SUBSTATION NAME	MAIN CLINT	LOCATION
Replacement of annunciator panel	Switching substation 13.8kv (29cc, 31cc and c1-1)	Saudi electricity	Jubail, KSA
Replacement of disconnecter, ground switch, transformer wall bushing and post insulator	Meridian substation, BOURSHED AND ANOOD 69/13.8KV SUSBTION	Saudi electricity	Khobar, KSA
Replacement and testing of Current transformer	Royal palace	Saudi electricity	Dammam
Full commissioning power transformers with replacement and testing	SAKAKA B, SAKAKA A, and DOUMAT	National grid company	SAKAKA, KSA
Replacement and testing of voltage transformer	Umm al sahiq 69/13.8kv substation	Saudi electricity	Umm al sahiq
Replacement and testing of current transformer	Dammam industrial#2 69/13.8kv substation	Saudi electricity	Dammam, KSA
Replacement and testing of wall bushing with CT	Khobar east 69/13.8kv substation	Saudi electricity	Khobar, KSA
Construction and testing of 380kv & 132kv OHTL interconnections of Layla BSP 9049	AL JOUF – 1 (9049 – 9067) OHL LENGTH – 216KM	HYUNDAI	AL JOUF, KSA
Replacement and testing of condenser busbar from transformer to switchgear	Sabic gas plant, Yanbu230 KV substation	Sabic gas plant	Yanbu, KSA
Repairing and testing of mobile transformer 30mva	Mobile transformer	National grid	Jubail. KSA
Replacement and testing of battery bank	C1-1 69/13.8kv substation	Saudi electricity	Jubail, KSA
Repairing and testing of power transformer 40mva	Khursaniyah 230kv substation	National grid	Khursaniyah ,ksa
Repairing and testing of power transformer 30mva	16bd, Jubail	National grid	Jubail, ksa
Installation and testing of power transformer 30mva	Qasumah power plant	Saudi electricity & national grid	Qasumah
Replacement and testing of battery charger	Faisaliyah, anak, midra al junubi sports city, meridian, Dammam bsp, manakh and airfield substation 69/13.8 kv substations	Saudi electricity	Dammam and Khobar area

132 KV / 220 KV SUBSTATION

- Extensive experience in the construction, installation, testing, commissioning, and energization of new 132 kV and 220 kV substations.
- Supervised and coordinated all electrical installation activities from project initiation through final handover.
- Managed installation, testing, and commissioning of Power Transformers, Auto Transformers, and Auxiliary Transformers.
- Supervised erection and commissioning of GIS (Gas Insulated Switchgear) and AIS (Air Insulated Switchgear) systems.
- Installation, testing, and commissioning of Circuit Breakers, Disconnectors (Isolators), Earthing Switches, and Current Transformers (CTs).
- Performed testing and commissioning of Voltage Transformers (VTs/CVTs), Surge Arresters, Neutral Grounding Resistors (NGR), and Busbar Protection Systems.
- Conducted insulation resistance, winding resistance, transformer turns ratio (TTR), magnetic balance, vector group verification, and capacitance/tan delta tests.
- Performed primary and secondary injection testing of protection and metering circuits.
- Experienced in testing and commissioning protection relays including Distance Protection, Differential Protection, Overcurrent Protection, Earth Fault Protection, Busbar Protection, and Transformer Protection.
- Configured, tested, and commissioned numerical protection relays from major manufacturers including Siemens, ABB, GE, Schneider, and SEL.
- Conducted end-to-end testing of protection schemes and interlocking systems.
- Performed SCADA, RTU, and communication system testing and integration with control centers.
- Experienced in substation automation systems (SAS), IEC 61850 communication protocols, and fiber optic networks.
- Supervised installation and testing of control and protection panels, marshalling kiosks, AC/DC distribution boards, battery chargers, and battery banks.
- Conducted cable laying, cable termination, cable glanding, cable tray installation, and cable route inspections.
- Performed HV cable testing including insulation resistance, sheath testing, continuity checks, and VLF testing.
- Conducted pre-commissioning checks, punch point clearance, and system readiness verification.
- Coordinated with clients, consultants, vendors, and utility representatives during testing, commissioning, and energization activities.
- Prepared commissioning procedures, method statements, test reports, punch lists, and final handover documentation.
- Participated in energization activities for transformers, switchgear, transmission line bays, and complete substation systems.
- Ensured compliance with IEC, IEEE, ANSI, Saudi Electricity Company (SEC), and project specifications.
- Maintained strict adherence to HSE requirements, permit-to-work systems, lockout/tagout (LOTO), and electrical safety procedures.
- Successfully completed SAT (Site Acceptance Test), FAT review, and final commissioning activities for substation projects.
- Experienced in troubleshooting and resolving protection, control, communication, and equipment-related issues during commissioning and operation.

- 1. Expertise in Cable Tray Installation:** Demonstrated proficiency in installing various types of cable trays, ensuring secure and organized cable management for complex electrical systems.
- 2. Cable Termination and Pulling:** Skilled in the precise termination and pulling of cables, with a strong emphasis on safety and minimizing damage or stress to cables, ensuring optimal connectivity and system reliability.
- 3. Power Cable Installation Specialist:** Experienced in the installation of power cables for a wide range of applications, from residential to industrial, focusing on efficient power distribution and adherence to electrical codes and standards.
- 4. Panel Installation Proficiency:** Detailed knowledge and practical experience in panel installation, including distribution boards and control panels, with an ability to troubleshoot and ensure seamless operation of electrical systems.

COMPANY NAME	ZAHARAN OPERATION AND MAINTENANCE COMPANY
DURATION	JUNE2017 TO JUNE 2022
PROJECT	BUILDING CONSTRUCTION OF PRINCE SULTAN MEDICAL CITY
LOCATION	RIYADH , KSA
DESIGNATION	ELECTRICAL SITE ENGINEER
COMPANY NAME	ZAHARAN OPERATION AND MAINTENANCE COMPANY

JOB DESCRIPTION

As an Electrical Site Engineer at Prince Sultan Medical City, I played a pivotal role in the construction and operational excellence of building projects within the medical city. My core responsibilities and achievements included:

➤ **Electrical Installation & Systems Management**

- Executed complete electrical installation activities, including cable trays, trucking, conduits, power and control cables, distribution boards, MCCs, PCCs, and electrical panels in accordance with project specifications and international standards.
- Supervised cable laying, cable pulling, cable dressing, and cable termination works for LV and MV systems, ensuring quality workmanship and timely project completion.
- Coordinated electrical construction activities and ensured compliance with approved drawings, technical specifications, and safety requirements.

➤ **AutoCAD & Technical Documentation**

- Prepared, reviewed, and updated electrical drawings using AutoCAD, including layout drawings, cable routing plans, single-line diagrams, and installation details.
- Managed drawing revisions and incorporated engineering changes to ensure accuracy and compliance with project requirements.
- Coordinated with consultants, clients, and engineering teams to resolve technical issues and drawing discrepancies.

➤ **Team Leadership & Site Supervision**

- Led and supervised electricians, technicians, and subcontractors during installation, testing, and commissioning activities.
- Provided technical guidance and training to site personnel, ensuring high productivity and quality standards.
- Monitored daily site activities, manpower allocation, and project progress to achieve project milestones within schedule.

➤ **Fire Alarm & HVAC Systems**

- Managed installation, testing, and commissioning support for fire alarm systems, ensuring compliance with safety codes and project specifications.
- Coordinated electrical works related to HVAC systems, including power supply, control wiring, and equipment connections.
- Ensured proper operation of ventilation, air conditioning, and life-safety systems within the facility.

➤ **Safety Management & Permit Compliance**

- Maintained strict adherence to HSE regulations, company safety policies, and client safety requirements.
- Conducted site inspections, toolbox talks, risk assessments, and safety audits to maintain a safe working environment.
- Worked as a Permit Receiver and ensured all electrical activities were carried out under approved work permits and safety procedures.
- Successfully maintained an excellent safety record by promoting a strong safety culture among the workforces.

➤ **Quality Assurance & Testing**

- Performed inspections, testing, and verification of electrical installations to ensure compliance with project specifications and international standards.
- Coordinated pre-commissioning and commissioning activities, ensuring all systems operated efficiently and reliably.
- Identified and resolved technical issues promptly, minimizing delays and improving system performance.

➤ **Coordination & Problem Solving**

- Collaborated closely with civil, mechanical, HVAC, instrumentation, and architectural teams to ensure smooth project execution.
- Demonstrated strong troubleshooting and problem-solving skills by identifying potential issues and implementing effective solutions.
- Maintained effective communication with clients, consultants, contractors, and project stakeholders throughout the project lifecycle.

➤ **Key Achievements**

- Successfully delivered electrical installation works for major commercial, industrial, and healthcare projects within budget and schedule.
- Led multidisciplinary teams and improved project execution efficiency through effective planning and supervision.
- Contributed to the successful installation and commissioning of fire alarm, HVAC, power distribution, and lighting systems.

- Achieved excellent safety performance with zero major incidents through strict implementation of HSE standards and procedures.
- Recognized for strong leadership, technical expertise, and commitment to quality and project success.

COMPANY NAME	AL MAWARID TRADING AND CONTRACTING
DURATION	2014 to 2017
PROJECT	MOI DATA CENTER
LOCATION	DOHA , QATAR
DESIGNATION	Electrical site engineer

MEP ELECTRICAL SITE ENGINEER | QATAR

Highly skilled MEP Electrical Site Engineer with extensive experience in the execution, supervision, installation, testing, commissioning, and maintenance of electrical systems for commercial, residential, industrial, and infrastructure projects. Proven expertise in managing electrical construction activities, ensuring compliance with project specifications, international standards, quality requirements, and HSE regulations.

KEY RESPONSIBILITIES

- Managed and supervised all electrical works related to MEP projects from project initiation through final handover.
- Coordinated site activities with consultants, clients, subcontractors, civil teams, mechanical teams, and project management personnel.
- Executed and supervised the construction of new electrical substations, including equipment foundations, cable trench systems, grounding systems, and electrical equipment installation.
- Supervised installation, testing, and commissioning of power transformers, distribution transformers, auxiliary transformers, and associated protection systems.
- Managed transformer room construction and ensured compliance with electrical safety standards, ventilation requirements, fire protection systems, and equipment clearances.
- Supervised installation, alignment, cable termination, and commissioning of diesel generators and generator synchronization systems.
- Coordinated installation and testing of Automatic Transfer Switch (ATS) panels and generator control systems.
- Supervised installation, testing, and energization of PCC (Power Control Center) panels, MCC (Motor Control Center) panels, MDBs, SMDBs, and distribution boards.
- Managed installation and termination of power, control, instrumentation, and communication cables.
- Supervised cable pulling, cable dressing, glanding, ferruling, termination, continuity testing, and insulation resistance testing for single-core, two-core, three-core, and four-core cables.
- Ensured proper segregation and routing of power and control cables according to approved engineering drawings.

- Supervised installation of cable trays, cable ladders, cable baskets, conduits, trunking systems, junction boxes, and underground duct banks.
- Conducted inspection of cable trenches and underground cable installations before backfilling activities.
- Managed installation of earthing and grounding systems including earth pits, earth mats, earth strips, and bonding networks.
- Supervised installation and testing of lightning protection systems in accordance with project specifications and international standards.
- Coordinated installation of lighting systems, emergency lighting systems, small power systems, and external lighting networks.
- Performed site inspections and quality control activities to ensure compliance with approved drawings and method statements.
- Reviewed electrical drawings, shop drawings, single-line diagrams, load schedules, cable schedules, and technical specifications.
- Prepared material take-offs, material requests, inspection requests, and work completion reports.
- Monitored project progress and ensured electrical works were completed according to project schedules and milestones.
- Conducted daily toolbox talks and ensured compliance with HSE requirements and permit-to-work procedures.
- Assisted in troubleshooting electrical faults and resolving technical issues during installation and commissioning stages.
- Participated in pre-commissioning, commissioning, functional testing, and system energization activities.
- Coordinated with utility authorities and consultants during inspections, testing, and final acceptance procedures.
- Prepared as-built drawings, operation manuals, testing reports, and final handover documentation.

EQUIPMENT AND SYSTEMS HANDLED

1. 11 kV / 33 kV Substations
2. Power Transformers
3. Distribution Transformers
4. Diesel Generators
5. ATS Panels
6. PCC Panels
7. MCC Panels
8. Main Distribution Boards (MDB)
9. Sub Main Distribution Boards (SMDB)
10. Capacitor Banks

- 11. Busbar Systems
- 12. Cable Trays and Cable Ladders
- 13. Earthing and Lightning Protection Systems
- 14. Transformer Rooms
- 15. Generator Rooms
- 16. Power and Control Cables (1-Core to 4-Core)
- 17. Lighting and Small Power Systems
- 18. Electrical Distribution Networks

MAJOR ACHIEVEMENTS

- Successfully completed installation and commissioning of new substations, transformer rooms, and generator rooms within project schedules.
- Executed large-scale cable installation and termination works while maintaining high-quality standards and zero major safety incidents.
- Successfully coordinated multidisciplinary teams to achieve timely project completion and client satisfaction.
- Ensured full compliance with international standards, project specifications, and HSE requirements throughout project execution

QUALIFICATION

- **M.TECH IN ELECTRICAL AND ELECTRONIS ENGINEERING FROM MAHARSHI DAYANAND UNIVERSITY, ROHTAK**
- **B.E IN ELECTRICAL & ELECTRONICS ENGINEERING 2013 FROM RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA BHOPAL**
- **INTERMEDIATE (SCIENCE) FROM BIHAR BOARD IN 2008**